

Programme: B E – CSE (AI&ML) and CSE (CS)
Internal Assessment – II

TERM: 03-03-2025 to 21-06-2025	COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS
DATE: 17/06/2025 TIME: 10:30am to 11:30am	COURSE CODE: CI43/CY43
MAX MARKS: 30	PORTIONS: L19 to L42
SEMESTER: 4	SECTION: 4A, 4B (AI&ML) and 4A (CS)



Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: 15x2=30

Q. NO	Questions	Blooms Levels	CO	Marks														
1.a	Differentiate between NP Hard and NP Complete Problems.	L2	CO5	5														
b	Consider the frequency table given in Table. 1b <table border="1"><thead><tr><th>Character</th><th>Frequency</th></tr></thead><tbody><tr><td>a</td><td>5</td></tr><tr><td>b</td><td>9</td></tr><tr><td>c</td><td>12</td></tr><tr><td>d</td><td>13</td></tr><tr><td>e</td><td>16</td></tr><tr><td>f</td><td>45</td></tr></tbody></table> Table. 1b i) Construct a Huffman Tree and compute the codeword for all the characters. ii) Encode “adadcabcbdaefabcde” using the codeword computed.	Character	Frequency	a	5	b	9	c	12	d	13	e	16	f	45	L3	CO3	5
Character	Frequency																	
a	5																	
b	9																	
c	12																	
d	13																	
e	16																	
f	45																	
c	Identify the minimum spanning tree for the graph given in Figure 1c. Illustrate the steps for computing the MST using Prim’s algorithm with source node as 1. Figure 1c.	L3	CO3	5														



2.a

Identify the maximum subset-sum and the maximum value that can be achieved for the following items with maximum capacity $W=6$. Apply dynamic programming algorithm to the given instance of knapsack problem in Table 2a.

Item	Weight	Profit
1	2	10
2	3	42
3	4	46

Table 2a.

L3

CO4

5

b

Find the shortest path from node 1 to every other node in Figure 2(b) graph using Bellman Ford Algorithm.

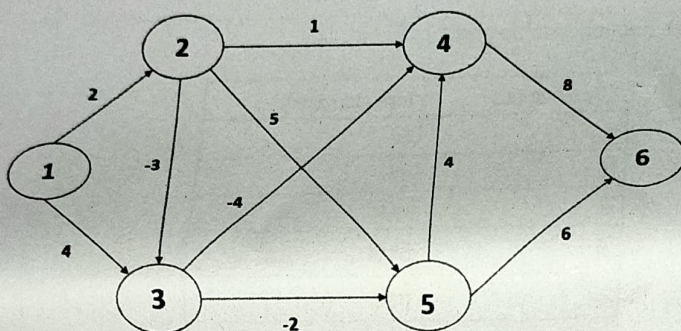


Fig.2(b) Graph

L3

CO4

5

c

Given the graph in Fig. 2(c), use Dijkstra's algorithm to find the shortest path from node A (START) to node G (END).

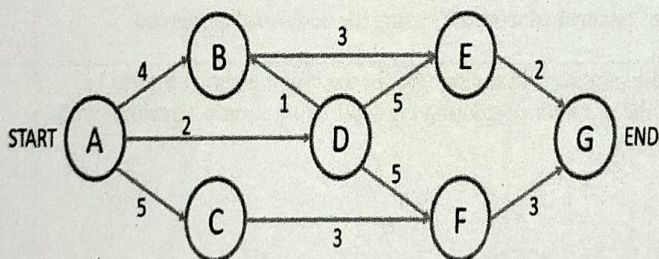


Fig. 2(c) Graph

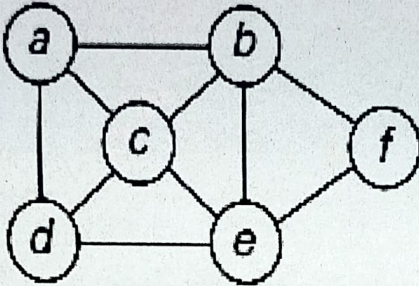
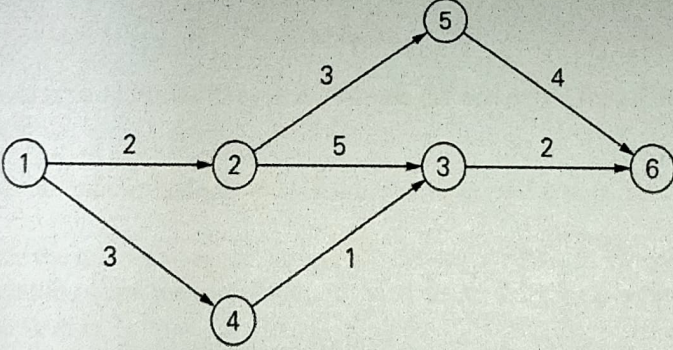
Compute the following:

- At each step, calculate the current shortest distances from A to all other nodes and represent it as a distance matrix
- In each iteration, show the node that is selected and compute the final shortest path from A to G along with the total cost of this shortest path.

L3

CO3

5

3.a	Solve the all pairs shortest-path problem using Flyod's Algorithm for the digraph with the weight matrix given in the graph in Fig 3(a) $\begin{pmatrix} 0 & 2 & \infty & 1 & 8 \\ 6 & 0 & 3 & 2 & \infty \\ \infty & \infty & 0 & 4 & \infty \\ \infty & \infty & 2 & 0 & 3 \\ 3 & \infty & \infty & \infty & 0 \end{pmatrix}$ Fig 3.a Graph	L3	CO4	5
b	Consider the graph given in Fig.3b. Apply backtracking to the given problem considering the start vertex as 'a'. Construct the State-space tree for finding a Hamiltonian circuit in the given graph.  Fig.3b.Graph	L3	CO5	5
c	Apply the Ford-Fulkerson's Algorithm to find a maximum flow and a minimum cut in the network given in Fig 3(c). Calculate the maximum flow pushed from source (1) to sink (6) through multiple augmenting paths by representing the residual graphs at each augmentation path.  Fig.3c Graph	L3	CO4	5

L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create